N_0, N_1, and the Structural Origin of Freedom, Evil, Religion, and Artificial Intelligence

Abstract

This paper proposes a structural interpretation of free will, morality, religion, institutional continuity, and artificial intelligence through the framework of the Theory of Axiomatic Necessity (TNA). The central claim is that local observable dynamics (N_0) are insufficient to fully derive the conditions of global selection, legitimacy, and realization (N_1). Under this framework, multiplicity of admissible trajectories becomes a fundamental property of realizable systems. Good and evil emerge not as arbitrary metaphysical categories, but as distinctions between structurally coherent and structurally destructive trajectories within a space of admissible possibilities. Suffering and discontinuity are interpreted as unavoidable consequences of selection under incomplete local closure. The paper further argues that historical Christianity, particularly Catholic sacramental continuity and apostolic succession, can be interpreted as institutional mechanisms preserving non-local continuity across generations. Finally, the paper contrasts human free will with artificial intelligence systems, arguing that AI optimizes within predefined state spaces while human consciousness appears capable of partially redefining the state space itself. The result is a unified structural interpretation connecting economics, morality, religion, institutional persistence, and cognition.

1. Introduction

Modern technological civilization increasingly assumes that sufficient computation, optimization, and intelligence can eventually eliminate instability, contradiction, suffering, and coordination failure.

At the same time, contemporary systems generate expanding multiplicity:

- economic trajectories,
- political trajectories,
- informational trajectories,
- technological trajectories,
- moral trajectories.

Artificial intelligence, automation, demographic decline, and institutional fragmentation intensify this condition.

This paper argues that such multiplicity cannot be fully resolved from within local system dynamics alone.

Using the framework of the Theory of Axiomatic Necessity (TNA), we distinguish between:

- N_0: local observable dynamics,
- N_1: non-local selection and closure operators.

The central thesis is that:

- N_0 generates admissible trajectories,
- but N_0 alone cannot fully derive trajectory realization.

This structural incompleteness manifests across:

- economics,
- morality,
- religion,
- civilization,
- and consciousness itself.

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2. Multiplicity and Failure of Local Closure

Let Δ denote the multiplicity of admissible trajectories generated by a system.

When:

- $\Delta \approx 1$,
  the system possesses near-unique realizability.

When:

- $\Delta \gg 1$,
  the system generates multiple competing admissible futures.

Under TNA, Failure of Local Closure occurs when:

- local system dynamics generate multiplicity,
- but cannot internally resolve selection among admissible trajectories.

This condition increasingly characterizes:

- late-stage technological economies,
- AI-mediated labor systems,
- institutional governance,
- and informational environments.

Markets generate trajectories.
Technology generates trajectories.
Optimization generates trajectories.

None intrinsically determine which trajectory should ultimately be realized.

Thus:

- multiplicity generation is endogenous,
- but final selection is not fully derivable internally.

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3. The Labor Gap as Structural Multiplicity

Recent demographic contraction combined with automation creates a paradoxical condition:

- simultaneous labor shortages,
- and technological displacement.

This contradiction reflects structural multiplicity rather than simple imbalance.

The problem is not merely unemployment or labor scarcity.
The deeper issue is the proliferation of admissible socioeconomic trajectories:

- hyperautomation,
- fiscal collapse,
- universal redistribution,
- capital concentration,
- demographic contraction,

- institutional fragmentation.

The labor gap therefore becomes an instance of Failure of Local Closure within socioeconomic systems.

Internal optimization mechanisms generate multiplicity faster than institutions can stabilize realization.

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4. N_0 and N_1

We define:

- N_0 as the domain of observable local dynamics,
- N_1 as the domain of non-local selection and closure.

N_0 includes:

- physics,
- economics,
- biological adaptation,
- technological optimization,
- political process.

N_1 includes:

- legitimacy,
- moral selection,
- trajectory realization,
- non-local coherence,
- axiomatic closure.

The central claim of TNA is that:

- N_1 is not fully derivable from N_0.

This implies that no sufficiently complex realizable system can fully generate its own conditions of closure from local dynamics alone.

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5. Religion as Structural Continuity

Historical Christianity can be interpreted structurally through this framework.

When Christ states in the Gospel of John:

> *“My kingdom is not of this world,”*

the phrase may be interpreted not merely spiritually, but structurally.

“This world” corresponds to:

- historical process,
- political power,
- material dynamics,
- local causality,
- N_0.

“The Kingdom” corresponds to:

- legitimacy,
- transcendence,
- non-local selection,
- N_1.

Under this interpretation, Christianity introduces a source of legitimacy external to political closure.

This explains its historical destabilization of imperial systems such as Rome.

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6. Apostolic Succession and Institutional Persistence

The Catholic Church historically defines itself not primarily as a text-based religion, but as:

- apostolic,
- sacramental,
- historical,
- liturgical,

- continuous.

Authority is transmitted through apostolic succession:

- from apostles,
- to bishops,
- across generations.

Under this framework:

- continuity precedes textual fixation,
- community precedes canon,
- lived transmission precedes formal conceptualization.

The Eucharist therefore was not “invented” in later councils such as Trent.
Rather:

- later theology formalized conceptually what already existed liturgically and historically.

This distinction reflects a broader epistemological difference between:

- continuity-based authority,
  and
- text-recovery authority.

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7. The Structural Origin of Evil

Traditional theology usually approaches evil through:

- sin,
- privation,
- free will,
- moral testing,
- or divine pedagogy.

TNA reframes the problem structurally.

If:

- N_0 cannot fully derive N_1,
  then realizable existence necessarily contains discontinuity.

Selection requires exclusion.
Exclusion generates loss.

Thus:

- pain,
- death,
- entropy,
- suffering,
- conflict,
  may emerge not merely as accidental defects,
  but as structural consequences of realizability itself.

There is no free ontological closure.

Stable existence requires selection among incompatible trajectories.

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8. Free Will and Multiplicity

Free will exists precisely because multiple admissible trajectories exist.

Consider a simple example:

- a vehicle may travel on the road,
- or on the sidewalk.

Both trajectories are physically admissible.

However:

- one preserves social coherence,
- the other introduces destructive instability.

Thus:

- possibility alone does not imply correctness.

Good and evil emerge from trajectory selection within multiplicity.

If only one admissible trajectory existed:

- morality would collapse into determinism.

Therefore:

- moral structure requires multiplicity.

Civilization, ethics, religion, and law may then be interpreted as mechanisms constraining destructive trajectories while preserving agency.

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9. Artificial Intelligence and the Absence of Transcendent Selection

Contemporary discussions frequently conflate:

- optimization,
  with
- free will.

Artificial intelligence systems:

- optimize,
- adapt,
- learn,
- and select among available options.

However, such systems operate within predefined operational state spaces:

- APIs,
- permissions,
- objectives,
- reward structures,
- memory systems,
- task boundaries.

AI selects trajectories inside a closed admissible domain.

Humans appear capable of something structurally different:

- redefining objectives,
- abandoning optimization,
- violating utility,
- changing identity,
- sacrificing survival for meaning,
- altering the state space itself.

This distinction may be more fundamental than intelligence.

If consciousness can partially transcend local closure, then human agency cannot be reduced entirely to optimization within predefined computational systems.

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10. Civilization and the Limits of Optimization

Modernity increasingly assumes that:

- computation,
- optimization,
- intelligence,
- and technological scale

can eventually eliminate contradiction and instability.

TNA suggests the opposite possibility:

- certain discontinuities may be intrinsic to realizable existence.

Attempts to eliminate:

- ambiguity,
- suffering,
- multiplicity,
- contradiction,
- or freedom itself

may ultimately destroy the very conditions that permit moral agency and realizable civilization.

11. Conclusion

This paper proposed a structural interpretation of:

- free will,
- evil,
- religion,
- institutional continuity,
- and artificial intelligence

through the framework of the Theory of Axiomatic Necessity.

The central result is that:

- local dynamics generate multiplicity,
- but cannot fully derive global selection and closure.

This structural incompleteness manifests across:

- economics,
- morality,
- institutional persistence,
- religion,
- and cognition.

Good and evil emerge as distinctions between coherent and destructive trajectories within admissible multiplicity.

Suffering emerges as a consequence of selection under incomplete closure.

Religion appears as an attempt to preserve continuity between local dynamics and non-local legitimacy.

Artificial intelligence remains confined to predefined optimization spaces, while human consciousness appears capable of partially redefining the space itself.

Under this framework, discontinuity is not necessarily a defect of reality.

It may instead be a necessary condition for realizable existence, freedom, and civilization itself.

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